Generative CAD for an Open-Source Multi-Powder Doser: A Tooling Step Toward Autonomous Discovery of Additively Manufactured Aerospace Alloys

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1 Introduction and Motivation

Additively manufactured (AM) structural alloys are an increasingly important class of materials for space applications, where launch-mass reduction, topology-optimized geometries, and rapid iteration on bespoke parts all favor laser powder-bed fusion (L-PBF) over conventional subtractive processes (TODO Author, 2024a). Discovering and qualifying *new* aerospace alloys for these processes – rather than only printing established compositions – is bottlenecked not by the printers themselves, but by the upstream workflow that prepares the metal powder feedstock from which each candidate composition is built.

A central, repetitive step in that workflow is *powder dosing*: weighing out and combining individual elemental or pre-alloyed powders into a target blend before each downstream alloy experiment. In a research lab this is almost always done by hand, one powder at a time, with a balance and a spatula. This is slow, error-prone, exposes researchers to fine reactive metal powders, and – critically – it caps the *compositional throughput* of any downstream alloy-discovery campaign.

Powder dispensing is therefore an *unmet need* in the research laboratory setting. The few commercial automated dispensers that do exist are priced for industrial QC labs and designed around pharmaceutical or analytical-chemistry workflows rather than aerospace alloy feedstock preparation, leaving them ill-matched to the volume-per-blend and particle-size regimes that L-PBF aerospace alloy work actually requires (TODO Author, 2026). Existing open-source designs, while encouraging, fall short of the rigorous design and testing needed for an L-PBF feedstock workflow, where the dispenser must respect aerospace-grade powder characteristics (e.g. tightly controlled particle-size distribution, sphericity, oxygen pickup) and must deliver total dispense volumes large enough to actually feed an alloy-screening pipeline (TODO Author, 2023).

This project will close that gap. We will use a mix of traditional mechanical design and *generative CAD* – driven primarily by agentic AI tooling layered on top of GitHub Copilot – with frequent build-test-feedback cycles between AI-generated designs and the human engineer to create a powder dosing system capable of handling **up to 30 unique powders** with **total dispense volumes up to 250 mL**, suitable as the front-end of an autonomous aerospace-alloy discovery loop.

The work has two complementary novel contributions:

1. A rigorous, documented exploration of the *capabilities and current limitations* of agentic AI applied to CAD and electromechanical design, using a real, non-trivial aerospace-relevant device as the test article.
2. An *open-source*, accurate, low-cost, reliable multi-powder dispenser purpose-built for L-PBF research workflows – released as a complete repository (CAD, firmware, BOM, build instructions, and characterization data) so other research groups can adopt and extend it.

A peer-reviewed paper documenting both contributions is planned as a follow-on deliverable.

2 Background

2.1 Powder dispensing in research and industry

Accurate dispensing of fine metal powders at research scale is dominated today by a small number of commercial systems aimed at analytical-chemistry and pharmaceutical applications. These systems achieve high single-dose accuracy, but at price points that are out of reach for most academic materials labs and with form factors and software stacks that assume a single feedstock at a time – exactly the opposite of what an alloy combinatorial campaign needs (TODO Author, 2026).

On the open-source / DIY side, prior work has demonstrated single-powder augers, vibratory dispensers, and small carousel feeders, but to our knowledge none of these published designs has been validated end-to-end against L-PBF feedstock requirements – particle-size-distribution preservation, cross-contamination between powders, oxygen exposure, and bulk volume per dispense – in the way an aerospace alloy discovery workflow demands (TODO Author, 2023). The result is a clear hole in the design space: a research-grade, low-cost, *multi-powder* dispenser sized for L-PBF feedstock volumes simply does not exist.

2.2 Generative CAD as a design accelerator

Generative CAD here refers to the broader family of techniques in which a designer specifies intent (functional requirements, constraints, interfaces) and software produces candidate geometry, rather than the designer drawing the geometry directly. In this project the "generator" is an agentic large-language-model loop – primarily GitHub Copilot backed by multiple model providers, augmented with task-specific tooling for CAD scripting, slicing, and printer control (see companion repository issues #6 and #24) – not a topology-optimization solver or a parametric template library, although those are complementary techniques the loop is free to invoke (TODO Author, 2025b,a).

The published research base on this style of LLM-driven, end-to-end electromechanical design is still thin: most of the literature is restricted either to topology optimization for a single fixed envelope or to short, demonstrative "can the model produce a bracket?" studies. The open question is whether a tightly-supervised agentic loop can deliver an *integrated* mechanical + electrical + firmware design for a real research instrument. This proposal directly probes that question.

2.3 Connection to autonomous AM alloy discovery

Autonomous, "self-driving" materials laboratories have shown the value of closing experimental loops with active learning over composition, process, and structure-property data (TODO Author, 2024b). Applied to AM aerospace alloys, the loop is gated by how quickly and reproducibly we can prepare new powder blends. An accurate, scriptable, 30-powder dispenser is the missing front-end for that loop and the direct technical contribution of this fellowship.

3 Project Objectives

1. **Open-source multi-powder doser (hardware).** A working prototype that handles up to 30 distinct powders with per-blend total dispense volumes up to 250 mL, with documented dispense accuracy, repeatability, and cross-contamination characterization on at least three representative L-PBF feedstock powders.
2. **Open-source repository.** A complete release at `github.com/vertical-cloud-lab/powder-doser` (Vertical Cloud Lab, 2026) containing CAD sources, firmware, BOM, assembly instructions, characterization data, and the AI prompts/transcripts that produced each design revision.

3. **Generative-CAD-first design process.** As little traditional, hand-driven CAD as we can manage – ideally the engineer should never need to open a CAD GUI to author or edit geometry, only to inspect and review what the agentic loop produced. Every deviation from that ideal will be logged and analyzed.
4. **Documented capability/limitation study.** A reproducible record of where agentic AI succeeded, where it failed, where engineer intervention was required, and what tooling unlocked each capability. This deliverable functions as a litmus test for the current state of generative CAD applied to a real electromechanical instrument and will be the basis of the follow-on paper.

It is important to state up front what this project is *not*: it is not an evaluation of AI's ability to develop a product unsupervised. A capable engineer (the applicant) will review the AI's contributions at every step and make meaningful corrections. The fellowship work is an honest evaluation of agentic AI as a *tool* that augments capable engineers, with generative CAD as the sharpest test case.

4 Approach and Methods

4.1 Generative CAD pipeline

The design loop is anchored on GitHub Copilot used in agent mode, with the freedom to call out to multiple underlying language models and to auxiliary tooling exposed through the project's repository (e.g. the component-selection work in repo issue #24 for the vibration motor and solenoid control stack on a Raspberry Pi Zero 2 W, and the meta-CAD tooling survey in #6). Each design iteration follows a build-test-feedback cycle:

1. The engineer describes a requirement or failure mode in plain language and reference data (test results, photos of failures, powder-flow observations).
2. The agentic loop produces or edits CAD sources, electrical layouts, and firmware – preferring scriptable, version-controlled formats so the diff is human-reviewable.
3. Prototypes of the dispenser are printed in-house on a Bambu Lab H2D, the printer chosen for this project's prototyping workflow. The long-term goal is for the agent to drive the printer *directly* using the H2D programmatic-print pathway being developed in repo issue #23, so the loop can iterate without an engineer-in-the-loop print step; in the near term the engineer triggers prints manually.
4. The engineer assembles, bench-tests, and reports results back into the loop. Each iteration's prompts, model outputs, and engineer corrections are committed to the repo so the design history is fully auditable.

4.2 Hardware design targets

The dispenser must, at minimum, satisfy the following functional requirements derived from the L-PBF feedstock-preparation use case (initial targets to be refined as bench data accumulate):

- Up to 30 distinct powder reservoirs, individually addressable.
- Total per-blend dispense volume up to 250 mL.
- Per-powder dispense accuracy compatible with compositional screening at the ± 1 wt% level for a 250 mL blend, with a stretch goal of ± 0.1 wt% for the smaller component powders that drive minor-alloying-element studies.
- Particle-size-distribution preservation across the dispense path (no significant attrition or segregation of the feedstock), verified by pre/post laser-diffraction PSD measurement.
- Negligible cross-contamination between adjacent powder reservoirs.

- Compatibility with reactive aerospace alloy powders. An inert-atmosphere enclosure is treated as a v2 deliverable; the v1 prototype targeted by this fellowship will be designed to accept such an enclosure but will be characterized in ambient conditions.

4.3 Validation and integration with the discovery loop

Validation combines bench-level tests of the dispenser in isolation (gravimetric repeatability over hundreds of doses, PSD measurement before/after, cross-contamination checks on at least three representative L-PBF feedstock powders) with an integration test in which the dispenser prepares blends consumed by the upstream end of our alloy-screening pipeline: the dosed powders are fed into *ultrasonic atomization* to produce spherical, L-PBF-grade feedstock, which is then printed and characterized. This places the dispenser directly on the critical path of the autonomous AM aerospace-alloy discovery loop and ensures the validation regime matches the real downstream use case. Results feed back into the generative-CAD loop both as design feedback and as the dataset that the follow-on paper will report.

5 Timeline and Deliverables

The bulk of the engineering work is targeted for the summer (May–August 2026), when the applicant has the most concentrated availability. The remainder of the calendar year is reserved for characterization, integration, and writing.

- **May 2026 (Months 0–1):** Finalize requirements; complete the issue #28 background sweep; stand up the agentic CAD loop end-to-end against a stub problem; select and procure mechanical and electrical components (cf. issue #24).
- **June 2026 (Month 2):** First full-system prototype with a ≤ 4-reservoir subset, printed on the H2D; basic dispense-accuracy bench tests; first round of generative-CAD capability/limitation observations logged.
- **July 2026 (Month 3):** Scale-up to the full 30-reservoir envelope; close the loop on direct printer control via issue #23 so the agent can iterate prints without engineer intervention; cross-contamination and PSD-preservation tests.
- **August 2026 (Month 4) – soft completion:** Integrated dispenser delivered, repository public-release-ready, capability/limitation study draft complete.
- **Fall 2026 (Months 5–7):** Integration test with the ultrasonic-atomization $\rightarrow$ L-PBF pipeline; expanded characterization; first paper draft.
- **Winter 2027 (Months 8–9):** Paper revisions and submission; community release of v1.0 of the open-source repository.

6 Broader Impact and Relevance to NASA

The most direct line to NASA is through aerospace structural alloys: a research-grade, low-cost, open-source multi-powder dispenser shortens the compositional-discovery loop for the L-PBF alloys that go into launch vehicles, in-space structures, and propulsion components. The same dispensing problem also appears, with different chemistries, in several other NASA-relevant areas: precise dispensing of fine powders for solid and hybrid rocket propellant formulation studies, regolith simulant handling for in-situ resource utilization (ISRU) experiments, and lunar/Martian regolith additive manufacturing research.

The agentic-AI portion of the work is also directly relevant to NASA's broader interest in AI-assisted engineering for cost- and schedule-constrained missions: a documented, honest assessment of where generative CAD currently helps and where it currently fails is exactly the kind of evidence that mission engineering teams need before they can adopt these tools. Finally, releasing the dispenser as a fully open-source design lowers the barrier of entry for other Utah and NASA-affiliated labs that want to run AM alloy-discovery campaigns without committing to industrial-scale capital equipment.

7 Qualifications of the Applicant

To be completed by the applicant.

References

TODO Author. PLACEHOLDER: Multi-powder dosing for compositional screening in laser powder-bed fusion (replace with real reference from issue #28). *TODO Journal*, 2023. Placeholder entry – replace with a peer-reviewed or well-cited open-source description of multi-powder dosing / blending hardware for L-PBF feedstock preparation.

TODO Author. PLACEHOLDER: Additive manufacturing of structural aerospace alloys – a review (replace with real reference from issue #28). *TODO Journal*, 2024a. Placeholder entry – replace with a peer-reviewed review of L-PBF aerospace alloys (e.g. Ti-6Al-4V, Inconel 718, Scalmalloy, GRX-810, etc.).

TODO Author. PLACEHOLDER: Self-driving laboratories for autonomous materials discovery (replace with a real reference, e.g. from the aspuru-guzik / berlinguette / nrc groups). In *TODO Venue*, 2024b. Placeholder entry – replace with a representative self-driving-lab reference.

TODO Author. PLACEHOLDER: Agentic large language models for electromechanical engineering tasks (replace with a real reference once one is identified). Pending issue #28 background sweep., 2025a. Placeholder entry – replace with a paper or technical report on LLM-agent-driven engineering / CAD design; literature here is thin, so a credible workshop paper or tech report is acceptable.

TODO Author. PLACEHOLDER: Generative computer-aided design for parametric mechanical assemblies (replace with real reference from issue #28). *TODO Journal*, 2025b. Placeholder entry – replace with a peer-reviewed reference on generative / AI-assisted CAD methodologies.

TODO Author. PLACEHOLDER: Landscape of commercial automated powder dispensing systems for laboratory use (replace with the outputs of issue #10). Pending issue #10 commercial-landscape sweep., 2026. Placeholder entry – once the issue #10 commercial sweep is complete, replace with the specific vendors / model numbers / price points we are positioning the open-source doser against.

Vertical Cloud Lab. `powder-doser`: open-source multi-powder dosing system (source repository). `https://github.com/vertical-cloud-lab/powder-doser`, 2026. Repository self-citation; safe to keep as-is.